

# MACHINE LEARNING

## COURSE MODALITIES AND MOTIVATION

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MASTER IN BUSINESS ANALYTICS



**UNIVERSITÉ  
DE GENÈVE**

# About myself

Sebastian Engelke

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- ▶ Born in Göttingen, Germany, where I did my PhD in mathematics/statistics.
- ▶ Worked at EPFL and HEC Lausanne since 2013.
- ▶ Joined the Research Center for Statistics in 2018.
- ▶ Research interests: extreme value theory; risk of meteorological rare events; spatial statistics; graphical models; machine learning; etc.  
⇒ [www.sengelke.com](http://www.sengelke.com)



# Course details

## Course Website

- ▶ On Moodle: <https://moodle.unige.ch/>
- ▶ Contains announcements, exercises, slides, data, code, etc.
- ▶ Course: “Machine Learning”; Password (later): ML2025

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## Exercises

- ▶ Thursday 16:15–18:00; SCIII 1S081 - Edouard Kellenberger
- ▶ Weekly assignments on moodle (not graded, but important material for exam)
- ▶ Assistants: Manuel Hentschel ([manuel.hentschel@unige.ch](mailto:manuel.hentschel@unige.ch))  
Christophe Valvason ([christophe.valvason@unige.ch](mailto:christophe.valvason@unige.ch))

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## Recordings

- ▶ The classes and seminar are recorded and available on Mediaserver
- ▶ Presence in person is strongly recommended to have the optimal learning experience and interactions with classmates

# Course assessment

## Final exam (70% of grade)

- ▶ 2-hour written exam during the official examination period
- ▶ Closed book
  - ▶ Allowed: a non-programmable calculator
- ▶ Prohibited material:
  - ▶ Everything else (e.g., notes, books, cell phones, laptops, etc.)



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## Data challenge (30% of grade)

- ▶ Best prediction on supervised learning task
- ▶ Deliverables: Jupyter notebook and written report
- ▶ Teams of 3 students allowed
- ▶ Start: week of October 27, 2025; End: December 21, 2025

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## Retake exam (100% of grade)

- ▶ During official retake exam period
- ▶ Same format and rules as for final exam
- ▶ In case of an unexcused absence or failed attempt of the final exam, the student can choose (before the retake) whether to keep the grade of the semester project or have the retake exam count for 100%.

### Main references

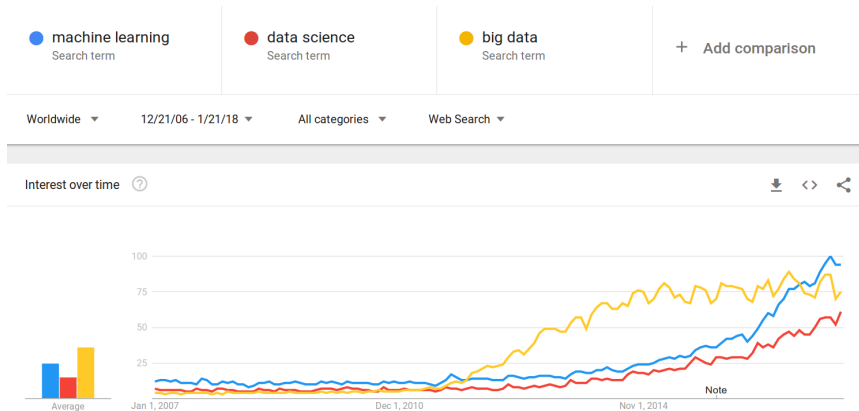
- [ISL] James, Witten, Hastie & Tibshirani. *An Introduction to Statistical Learning*, Springer.  
Free pdf on <http://www-bcf.usc.edu/~gareth/ISL/>
  - [ESL] Hastie, Tibshirani & Friedman. *The Elements of Statistical Learning*, Springer.  
Free pdf on <http://web.stanford.edu/~hastie/ElemStatLearn/>
  - [HOM] Géron. *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media, Inc.
- ▶ Several figures and data sets are taken from these books.

### Some related websites

- ▶ Machine learning competition: <https://www.data-mining-cup.com/>
- ▶ More competitions: <https://www.kaggle.com/>
- ▶ News on machine learning related topics: <https://www.wired.com/>

# Machine learning is trendy

Google Trends: interest in buzz words 2007–2017.



# Best job in 2017

Glassdoor, best jobs in America in 2017.

## 1 Data Scientist



**4.8 / 5**  
Job Score

**4.4 / 5**  
Job Satisfaction

**\$110,000**  
Median Base Salary

**4,184**  
Job Openings

[View Jobs](#)

## Culture: AlphaGo

*After humanity spent thousands of years improving our tactics, computers tell us that humans are completely wrong... I would go as far as to say not a single human has touched the edge of the truth of Go. [Ke Jie; Go Champion]*



**Figure:** In 2015, the first computer program won against a professional Go player, based on a combination of deep neural nets and reinforcement learning; see

<https://www.deepmind.com/research/highlighted-research/alphago>

## Sociology: artificial intelligence



## Technology: smart cars

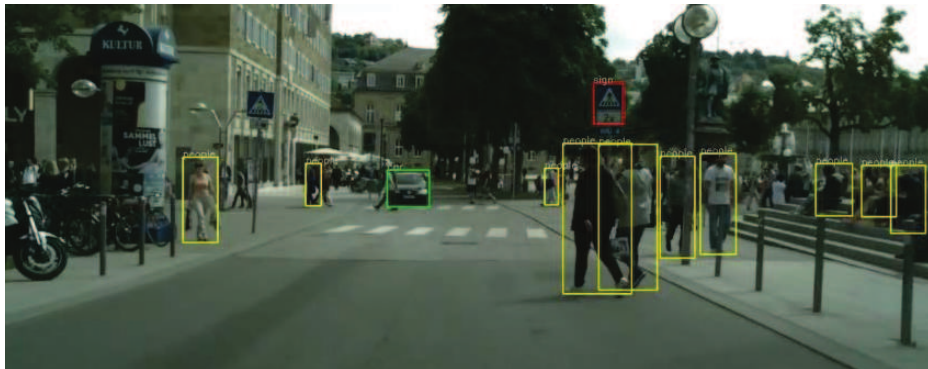


Figure: CES 2016: NVIDIA DRIVENet Demo - Visualizing a Self-Driving Future; see [www.youtube.com/watch?v=HJ58dbd5g8g](http://www.youtube.com/watch?v=HJ58dbd5g8g)



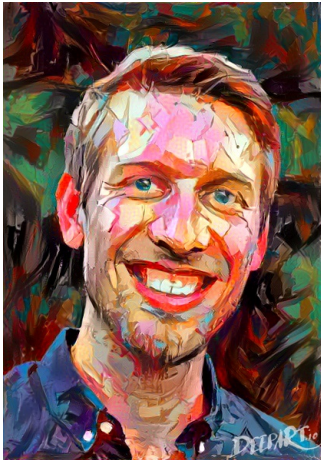




Art: <https://deepart.io/>



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# Applications of machine learning

## **Business**

- ▶ Work flows in businesses rely heavily on computer programs based on machine learning, e.g., market forecasting, customer segmentation and retention, fraud detection, etc.
- ▶ Most start-ups today develop products/technology based on machine learning.

## **Data mining and new technologies**

- ▶ Self-customizing programs: e.g., Amazon, Netflix, recommender systems
- ▶ Computer vision, autonomous driving
- ▶ Language processing: ChatGPT

## **Science**

- ▶ Applications in medicine such as disease diagnostics based on medical records
- ▶ Environmental applications such as weather forecasting

## **Artificial intelligence**

- ▶ Robot navigation
- ▶ Game AI: chess, Go, computer games, etc.